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Project Report : Queue App

1. Introduction

the app i made is a queue management app for the Algerian post departement (La Poste Algerienne). the idea is simple, instead of going to the post office and waiting for a long time not knowing when your turn will come, you can just reserve your ticket from your phone. this gives everyone a fair chance to pass one after one and saves alot of time.

i choosed this idea because its a real problem that algerians face everyday in the post office. the queues are long and disorganized so i thought it would be a good idea to solve it with a mobile app.

2. Application Screens

2.1 Home Screen

this is the entry point of the application. it shows the name of the app and a list of services that La Poste Algerienne provides. the user can choose which service he wants like sending a package, CCP withdrawal, paying bills, or other services.

2.2 Ticket Screen

this screen comes after the user selects a service. it shows the ticket that was given to the user with all the informations he needs :

- his ticket number (example : A-47)
- the service type he chose
- how many people are before him in the queue
- the estimated time before his turn

2.3 Queue Screen

this screen shows the current state of the waiting room. it displays all the tickets in the queue and shows which ticket is currently being served. the user can also see where his ticket is standing in the list. his ticket is highlighted so he can find it easily.

3. Components Used

here is the list of Jetpack Compose components i used in this project :

- Scaffold : used in all screens to provide the basic screen structure with a top bar
- TopAppBar : the top bar showing the title of each screen
- LazyColumn : used to display the list of services and the queue list in a performant way

- Card : used to display each service, ticket info, and queue item
- Column : to arrange elements vertically
- Row : to arrange elements horizontally side by side
- Text : to display all text in the app
- Spacer : to add space between elements
- Modifier : to style the components (padding, size, colors...)

4. Difficulties

the main difficulty i faced was understanding how LazyColumn works, specifically the difference between item() and items(). item() is used for a single element and items() is used for a list. also i had to understand how LazyColumn only renders the visible elements on screen which makes it more optimized than a normal Column.

another thing i found interesting is that Kotlin has inline if statements similar to Rust, for example you can write : color = if (ticket == myTicket) Color.Blue else Color.Gray directly inside a parameter without needing a separate variable.

i also got a warning about TopAppBar being experimental in Material 3, i fixed it by adding @OptIn(ExperimentalMaterial3Api::class) above the function.

5. Conclusion

this project helped me understand the basics of Jetpack Compose and how to build a real Android UI. i learned how to use layouts like Column, Row and LazyColumn to organize the interface, and how to use components like Card and Scaffold to make it look professional.

the app is not fully functional but the interface is complete with 3 screens that represent a realistic queue management system for La Poste Algerienne.